

```
[99]: # 🏠 Step 8 – Save the Entire Prediction Machine
      # We'll use joblib, which is commonly used to save scikit-learn models.
```

```
[100]: import joblib
      joblib.dump(final_model, "california_housing_model.pkl")
```

```
[100]: ['california_housing_model.pkl']
```

```
[101]: # Confirm the file exists
```

```
[102]: import os

      print(os.path.exists("california_housing_model.pkl"))

      True
```

```
[103]: # we can load the saved machine without rebuilding everything:
      loaded_model = joblib.load("california_housing_model.pkl")
```



Step Give Our Model a NEW House

```
[105]: # Create a new house
```

```
[106]: new_house = X_train.iloc[[0]].copy()
```

```
[107]: new_house["longitude"] = -118.25
      new_house["latitude"] = 34.05
      new_house["housing_median_age"] = 25
      new_house["total_rooms"] = 2500
      new_house["total_bedrooms"] = 500
      new_house["population"] = 1200
      new_house["households"] = 450
      new_house["median_income"] = 6.5
      new_house["ocean_proximity"] = "NEAR OCEAN"
```



Step Ask the prediction machine

```
[108]: prediction = final_model.predict(new_house)
      print("Predicted median house value:", prediction[0])
```

Predicted median house value: 263729.0



Step Get a Clean, Human-Readable Prediction

```
[109]: predicted_price = prediction[0]

      print(f"Predicted median house value: ${predicted_price:,.0f}")
```

Predicted median house value: \$263,729



Step Build Our Prediction Interface

```
[111]: # Since you're working in a notebook, we'll use Gradio for a clean interactive interface.
      # Install Gradio
```

```
[118]: import gradio as gr
      import pandas as pd
```

```
[119]: # Create the prediction function
```

```
[120]: def predict_house_value(
    longitude,
    latitude,
    housing_median_age,
    total_rooms,
    total_bedrooms,
    population,
    households,
    median_income,
    ocean_proximity
):

    # Feature engineering
    rooms_per_household = total_rooms / households
    bedrooms_per_room = total_bedrooms / total_rooms
    population_per_household = population / households
```

```
# Create raw input DataFrame
new_house = pd.DataFrame([
    "longitude": [longitude],
    "latitude": [latitude],
    "housing_median_age": [housing_median_age],
    "total_rooms": [total_rooms],
    "total_bedrooms": [total_bedrooms],
    "population": [population],
    "households": [households],
    "median_income": [median_income],
    "rooms_per_household": [rooms_per_household],
    "bedrooms_per_room": [bedrooms_per_room],
    "population_per_household": [population_per_household],
    "ocean_proximity": [ocean_proximity]
])

# Prediction
prediction = final_model.predict(new_house)[0]

return f"${prediction:,.0f}"
```



Step — Build the interface

```
[121]: interface = gr.Interface(
    fn=predict_house_value,

    inputs=[
        gr.Number(label="Longitude"),
        gr.Number(label="Latitude"),
        gr.Number(label="Housing Median Age"),
        gr.Number(label="Total Rooms"),
        gr.Number(label="Total Bedrooms"),
        gr.Number(label="Population"),
        gr.Number(label="Households"),
        gr.Number(label="Median Income"),
        gr.Dropdown(
            choices=[
                "<1H OCEAN",
                "INLAND",
                "ISLAND",
                "NEAR BAY",
                "NEAR OCEAN"
            ],
            label="Ocean Proximity"
        )
    ],

    outputs=gr.Textbox(label="Predicted Median House Value"),

    title="🏠 California House Value Predictor",

    description="Enter house features and let our Random Forest model estimate the median house value."
)
```

And finally launching the model interface

```
[122]: interface.launch()

* Running on local URL: http://127.0.0.1:7860
* To create a public link, set `share=True` in `launch()`.
```



California House Value Predictor

AI-powered median house-value estimation

Enter the characteristics of a California housing district and let our trained **Random Forest** model estimate its median house value.

📍 Location

Longitude

Latitude

🏠 Property

Housing Median Age

Total Rooms

Total Bedrooms

👤 Population & Households

📊 Income & Location Type

Population	Median Income
1200	6.5
Households	Ocean Proximity
450	NEAR OCEAN

Predict House Value

Prediction will appear here

Model: Random Forest Regression

R²: 0.8067

The prediction is an ML estimate based on patterns learned from the California Housing dataset. It is not a guaranteed market valuation.

Use via API · Built with Gradio · Settings

```
[123]: # Create the polished app skeleton

[127]: import gradio as gr

with gr.Blocks(title="California House Value Predictor") as app:

    gr.Markdown(
        """
        # 🏠 California House Value Predictor
        ### AI-powered median house-value estimation

        Enter the characteristics of a California housing district
        and let our trained **Random Forest** model estimate its
        median house value.
        """
    )

    with gr.Row():

        with gr.Column():
            gr.Markdown("### 📍 Location")

            longitude = gr.Number(
                label="Longitude",
                value=-118.25
            )

            latitude = gr.Number(
                label="Latitude",
                value=34.05
            )
```

```
with gr.Column():
    gr.Markdown("### 🏠 Property")

    housing_median_age = gr.Number(
        label="Housing Median Age",
        value=25
    )

    total_rooms = gr.Number(
        label="Total Rooms",
        value=2500
    )

    total_bedrooms = gr.Number(
        label="Total Bedrooms",
        value=500
    )

with gr.Row():

    with gr.Column():
        gr.Markdown("### 👤 Population & Households")

        population = gr.Number(
            label="Population",
            value=1200
        )

        households = gr.Number(
            label="Households",
            value=450
        )
```

```

with gr.Column():
    gr.Markdown("### 🏠 Income & Location Type")

    median_income = gr.Number(
        label="Median Income",
        value=6.5
    )

    ocean_proximity = gr.Dropdown(
        choices=[
            "<1H OCEAN",
            "INLAND",
            "ISLAND",
            "NEAR BAY",
            "NEAR OCEAN"
        ],
        value="NEAR OCEAN",
        label="Ocean Proximity"
    )

predict_button = gr.Button(
    "🔮 Predict House Value",
    variant="primary"
)

prediction_output = gr.Markdown(
    "### 📊 Prediction will appear here"
)

```

```

gr.Markdown(
    """
    ---
    **Model:** Random Forest Regression
    **R2:

```

```

# 🚀 Launch the application
app.launch()

```

```

* Running on local URL: http://127.0.0.1:7861
* To create a public link, set `share=True` in `launch()`.

```